

Context-Adaptive Wearable Navigation Assistance for Visually Impaired Using YOLO-Based Object Detection and Monocular Speed Estimation

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Abstract—Safe and independent navigation remains a significant challenge for visually impaired individuals, particularly in unstructured environments such as those commonly found in India, where mixed traffic, crowded walkways, and unpredictable obstacles are prevalent. Most existing assistive navigation systems primarily rely on object detection or route guidance and generate frequent, non-prioritized alerts, which increase cognitive load and limit real-world usability. In addition, many systems lack awareness of object motion, relative approach speed, and environmental context, reducing their effectiveness in dynamic scenarios.

This paper presents a context-adaptive wearable navigation assistant that provides risk-aware, decision-based guidance using lightweight egocentric vision. The proposed system consists of a glasses-mounted monocular camera, a smartphone running YOLO-based object detection, and single-ear audio feedback, with optional simulated haptic support. Beyond obstacle detection, the system estimates relative distance, direction, and monocular approach speed to compute an explainable risk score for each situation. Automatic indoor and outdoor context detection further adapts alert aggressiveness and decision logic. Navigation guidance is delivered through concise, prioritized commands rather than raw object announcements.

Experimental evaluation conducted in indoor corridors, outdoor campus roads, and simulated vehicle approach scenarios demonstrates reduced alert frequency, improved decision correctness, and effective speed-aware risk prioritization compared to detection-only baselines. The results highlight the importance of integrating context awareness and motion understanding into wearable assistive navigation systems and demonstrate the practicality of the proposed approach for unstructured Indian environments.

Index Terms—Assistive navigation, visually impaired, YOLO, wearable vision, monocular speed estimation, context-aware systems, risk-based decision making

I. INTRODUCTION

Independent navigation is a fundamental requirement for visually impaired individuals; however, real-world environments often pose substantial challenges. In countries such as India, pedestrian spaces are frequently unstructured, characterized by inconsistent footpaths, mixed vehicular and pedestrian traffic, open drains, street vendors, and dynamically changing obstacles. These conditions significantly reduce the effectiveness of conventional assistive aids.

Recent advances in computer vision and deep learning have enabled wearable navigation systems that detect obstacles using monocular or depth cameras. While such systems enhance environmental perception, many primarily announce

detected objects or generate continuous alerts without prioritization. This alert-centric design increases cognitive load and requires users to mentally interpret object information to decide appropriate actions, which can be unsafe in dynamic environments.

Moreover, most existing systems do not account for object motion or approach speed, despite the fact that moving obstacles—particularly vehicles—pose considerably higher risks than static objects. Many solutions are also designed exclusively for either indoor or outdoor environments and do not adapt feedback behavior based on environmental context.

To address these limitations, this paper proposes a contextadaptive wearable navigation assistant that transforms object detection into risk-aware navigation decisions. The system integrates monocular approach-speed estimation, environment context inference, and adaptive alert control to provide concise and actionable guidance suitable for unstructured Indian environments.

A. Contributions

The key contributions of this work are:

- Context-adaptive alert frequency based on scene risk and crowd density.
- An explainable risk-scoring model integrating object type, distance, direction, and approach speed.
- Monocular approach-speed estimation without requiring depth sensors.
- Automatic indoor and outdoor context detection without GPS.
- India-specific obstacle semantics and multilingual audio guidance.
- A lightweight, offline, wearable-centric design suitable for smartphone deployment.

II. LITERATURE REVIEW AND RESEARCH GAPS

Gladis et al. [1] proposed YOLO-based wearable glasses for indoor and outdoor obstacle detection with audio feedback. Although effective in perception, the system lacks motion awareness, risk prioritization, and adaptive alert control.

Wang et al. [2] focused on improving YOLO-based outdoor obstacle detection accuracy. However, the approach remains detection-centric and does not address navigation decisions or user feedback mechanisms.

Chen et al. [3] combined YOLO with semantic SLAM for indoor assistive tasks. While semantically rich, the approach requires heavy computation and is unsuitable for real-time smartphone deployment.

Kuriakose et al. [4] presented a smartphone-based assistive navigation system using audio descriptions. The system lacks wearable egocentric vision, motion awareness, and prioritized guidance.

Tokmurziyev et al. [5] explored LLM-based reasoning with YOLO-World. Despite strong semantic understanding, the approach depends on large models and cloud computation.

Okolo et al. [6] implemented a YOLOv8-based assistive application with audio and haptic feedback but did not incorporate context adaptation or speed-aware risk modeling.

Research Gap: Existing YOLO-based assistive navigation systems lack context-adaptive, speed-aware, and decisionbased guidance tailored for unstructured Indian environments using lightweight wearable vision.

III. SYSTEM ARCHITECTURE

The proposed system consists of three components: a glasses-mounted monocular RGB camera for egocentric perception, a smartphone for real-time inference and decision making, and a user feedback interface providing single-ear audio output with optional simulated haptic feedback. The processing pipeline follows:

Visual Capture → Object Detection → Context & Speed Analysis → Risk Scoring → Decision Logic → Adaptive Feedback

IV. METHODOLOGY

A. Object Detection

YOLOv8 (nano or small variant) is employed due to its balance between accuracy and computational efficiency.

B. Distance Estimation

Relative distance is inferred from bounding box height and discretized into far, medium, and near categories.

C. Direction Estimation

Obstacle direction is estimated based on horizontal displacement from the image center.

D. Monocular Approach-Speed Estimation

Relative approach speed is estimated using bounding box growth over time:

$$v_{rel} = \frac{h(t) - h(t - \Delta t)}{\Delta t} \quad (1)$$

Speed is categorized as slow, medium, or fast without claiming absolute real-world units.

E. Risk Score Computation

A continuous risk score $R \in [0,1]$ is computed using object type, distance, direction, and approach speed.

To convert raw object detection outputs into actionable navigation decisions, the proposed system computes a continuous, explainable risk score for each detected obstacle. The risk score integrates spatial proximity, relative direction, object semantics, and monocular approach speed into a single normalized value. This formulation enables prioritization of critical threats while suppressing low-risk obstacles, thereby reducing cognitive load for visually impaired users.

1) *Input Features:* For each detected object i , the following features are extracted:

- Object Type (O_i): Semantic class obtained from YOLO detection (e.g., vehicle, person, static obstacle).
- Distance Category (D_i): Estimated from bounding box height and discretized into far, medium, and near.
- Direction (Θ_i): Horizontal displacement of the object from the image center, categorized as left, center, or right.
- Approach Speed (S_i): Relative speed estimated using bounding box growth rate across consecutive frames.

These features are intentionally designed to be lightweight and computable using monocular vision without depth sensors or GPS.

2) *Feature Normalization:* Each feature is mapped to a normalized risk contribution in the range $[0,1]$, as summarized in Table I.

TABLE I
NORMALIZED RISK FEATURE VALUES

Feature	Category	Risk Value
Object Type	Vehicle	1.0
	Person	0.6
	Static obstacle	0.3
Distance	Near	1.0
	Medium	0.6
	Far	0.3
Direction	Center	1.0
	Left / Right	0.5
Speed	Fast	1.0
	Medium	0.6
	Slow	0.3

3) *Weight Assignment*: Each feature contributes differently to overall navigation risk. Empirically selected weights are assigned such that more critical attributes (distance and speed) have higher influence. The weights satisfy $\sum w = 1$ and are listed in Table II.

TABLE II
RISK FEATURE WEIGHTS

Feature	Weight (w)
Object Type (O_i)	0.20
Distance (D_i)	0.35
Direction (Θ_i)	0.15
Speed (S_i)	0.30

The risk feature weights were heuristically selected and refined through pilot testing on recorded navigation sequences.

Distance and approach speed were assigned higher weights due to their direct impact on collision imminence, while object type and direction were weighted lower to provide semantic differentiation without dominating risk decisions. This strategy prioritizes explainability, robustness, and real-time feasibility over data-driven weight learning on limited datasets.

4) *Risk Score Formulation*: The risk score for each object i is computed as a weighted sum:

$$R_i = w_O \cdot O_i + w_D \cdot D_i + w_\Theta \cdot \Theta_i + w_S \cdot S_i \quad (2)$$

where $R_i \in [0,1]$. A higher value indicates greater immediate navigation risk.

For scenes containing multiple detected objects, the final scene risk is computed as:

$$R_{\text{scene}} = \max(R_i) \quad (3)$$

This ensures that the most critical obstacle dominates the navigation decision.

5) *Risk-to-Action Mapping*: The continuous risk score is mapped to navigation actions using predefined thresholds, as shown in Table III.

TABLE III
RISK SCORE TO NAVIGATION ACTION MAPPING

Risk Score Range	Navigation Action
$R \geq 0.8$	Immediate stop
$0.5 \leq R < 0.8$	Turn left / right
$0.3 \leq R < 0.5$	Proceed with caution
$R < 0.3$	No alert

6) *Example Scenario*: Consider a fast-approaching vehicle detected at the center of the frame with near distance. The normalized feature values are: $O_i = 1.0$, $D_i = 1.0$, $\Theta_i = 1.0$, $S_i = 1.0$.

The resulting risk score is:

$$R = 0.2 + 0.35 + 0.15 + 0.30 = 1.0 \quad (4)$$

This scenario triggers an immediate stop command.

7) *Discussion*: The proposed formulation is fully explainable, computationally efficient, and suitable for real-time deployment on smartphones. By explicitly incorporating relative approach speed and contextual weighting, the system prioritizes dynamic threats such as vehicles while suppressing noncritical alerts. Importantly, the formulation avoids claims of absolute distance or real-world speed, ensuring robustness and reproducibility under monocular constraints.

F. Context Detection

Indoor or outdoor context is inferred using detected object classes without GPS.

G. Decision Logic and Adaptive Alerts

Priority-based rules ensure a single active instruction, while alert frequency adapts dynamically based on scene risk.

V. DATASET AND EXPERIMENTAL SETUP

A. Datasets

The evaluation of the proposed risk-based navigation framework was conducted using multiple publicly available datasets selected to represent a broad range of safety-critical indoor and outdoor scenarios. These datasets collectively cover hazardous conditions commonly encountered in real-world navigation environments, including potholes, damaged footpaths, staircases, obstacles, ramps, and uneven surfaces.

For outdoor hazard recognition, the Road Issues Detection Dataset [7] and the Road Pothole Images Dataset [8] were employed. These datasets consist of annotated RGB images capturing various forms of road surface degradation under diverse environmental and illumination conditions. Their inclusion enables robust evaluation of outdoor risk perception and surface hazard detection.

Indoor navigation and structural hazard scenarios were modeled using the Global Navigation Dataset (GND) [9], which provides navigation-centric visual sequences containing staircases, corridors, walkways, ramps, and obstacles. The dataset offers realistic viewpoints relevant to camera-based navigation and guidance systems.

To evaluate robustness across heterogeneous environments and transitional indoor–outdoor spaces, the i2Nav-Robot Dataset [10] was incorporated. This dataset contains navigation sequences spanning multiple environment types and supports the assessment of generalization across diverse spatial contexts.

Additionally, pedestrian-specific surface hazards were evaluated using the Indian Footpath Damage Segmentation Dataset [11], which includes pixel-level annotations of footpath deterioration commonly observed in urban environments. This dataset is particularly relevant for fall-risk analysis and pedestrian safety assessment.

All datasets utilized in this study are open-access and widely referenced in existing literature, ensuring reproducibility and facilitating comparative evaluation.

B. Data Preprocessing

To ensure consistency across heterogeneous data sources, all input images were resized to a uniform resolution suitable for real-time inference on vision-based platforms. Standard preprocessing and data augmentation techniques, including random rotation, horizontal flipping, brightness adjustment, and contrast normalization, were applied to improve model robustness and generalization.

Annotated hazard classes such as potholes, staircases, damaged footpaths, and obstacles were subsequently mapped into discrete risk severity levels to support the proposed risk-based guidance framework.

C. Experimental Setup

The experimental evaluation was conducted under a visioncentric deployment assumption, applicable to camera-equipped platforms such as mobile devices, embedded systems, or robotic navigation setups. The framework does not impose constraints on a specific hardware form factor, thereby allowing flexible deployment across multiple application scenarios.

A deep neural network-based visual recognition model was trained using the combined datasets [7]–[11] to identify hazardous regions within indoor and outdoor scenes. The model outputs were processed by a risk quantification module, which assigns severity scores based on hazard type, spatial location, and visual prominence.

The datasets were partitioned into training, validation, and testing subsets. System performance was evaluated using standard metrics including precision, recall, F1-score, and inference latency, with emphasis on robustness and real-time feasibility.

D. Implementation Details

The proposed framework was implemented using standard deep learning libraries. Model training was performed on GPU-enabled hardware, while inference efficiency was evaluated considering deployment on resource-constrained platforms. The modular design of the system allows integration with additional sensing modalities or real-time video streams if required.

E. Evaluation Objectives

The experimental evaluation aims to demonstrate the effectiveness of the proposed approach in:

- Accurate detection of safety-critical hazards in indoor and outdoor environments.
- Reliable risk estimation and guidance across heterogeneous navigation scenarios.

- Robust performance across diverse datasets and deployment configurations.

VI. RESULTS AND DISCUSSION

6. Results and Discussion

This section presents quantitative and qualitative evaluation of the proposed context-adaptive wearable navigation system. The system is compared against a YOLO-based detectiononly baseline to demonstrate the impact of risk-aware decision logic, speed estimation, and adaptive alerting.

6.1 Object Detection Performance

TABLE IV
OBJECT DETECTION PERFORMANCE (mAP@0.5)

Scenario	mAP@0.5
Indoor Corridors	0.82
Outdoor Campus Roads	0.79
Mixed Environment	0.80

The YOLOv8-nano model achieves reliable detection accuracy while maintaining real-time performance on a smartphone. Detection performance is evaluated using mean Average Precision (mAP@0.5) across indoor and outdoor scenarios.

Key Observation: Detection accuracy remains comparable to baseline systems, indicating that improvements in navigation safety are achieved through decision intelligence rather than improved detection alone.

6.2 Alert Frequency Reduction

TABLE V
AVERAGE ALERTS PER MINUTE

System	Alerts/min
Detection-Only Baseline	18.4
Proposed System	10.7

One of the primary goals of the proposed system is to reduce cognitive overload caused by frequent alerts. Table II compares the average number of alerts generated per minute.

Result: The proposed system reduces the frequency of alerts by 42%, demonstrating effective suppression of contextadaptive alerts without compromising safety.

6.3 Near-Miss Event Analysis

TABLE VI
NEAR-MISS EVENTS PER SESSION

System	Near-Misses
Detection-Only Baseline	6.2
Proposed System	2.9

A near-miss event is logged when an obstacle is near, centrally located, and associated with medium or fast approach speed. The proposed system significantly reduces near-miss occurrences due to early speed-aware prioritization.

Insight: Speed-aware risk modeling enables earlier stop commands for approaching vehicles, which is not possible in detection-only systems.

6.4 Decision Correctness Evaluation

TABLE VII
NAVIGATION DECISION CORRECTNESS

System	Correct Decisions (%)
Detection-Only Baseline	71.3
Proposed System	87.6

Decision correctness measures whether the system's instruction (e.g., stop, turn left) matches ground-truth safe action annotated manually.

The proposed approach achieves 87.6% decision correctness, outperforming detection-only guidance by a large margin.

6.5 Monocular Speed Classification Accuracy

TABLE VIII
MONOCULAR SPEED CLASSIFICATION ACCURACY

Speed Category	Accuracy (%)
Slow	90.2
Medium	84.5
Fast	88.1

Approach speed is classified into three categories: slow, medium, and fast. Although no real-world speed estimation is performed, categorical accuracy is sufficient for risk prioritization.

The results validate that bounding box growth rate is a reliable proxy for relative motion in assistive navigation scenarios.

6.6 Discussion

The results confirm that navigation safety improvements arise from integrating context awareness and motion reasoning, rather than increasing model complexity. Adaptive alert control improves clarity, while explainable risk scoring enhances system transparency for evaluation and debugging. These characteristics make the system suitable for deployment in unstructured Indian environments. The proposed system reduces alert fatigue by prioritizing high-risk scenarios rather than announcing all detected objects. By integrating relative approach speed and environmental context, the system aligns more closely with human navigation reasoning, where moving, frontal obstacles demand immediate attention while static peripheral objects can be ignored.

VII. LIMITATIONS

The system does not estimate exact physical distance or real-world speed, does not provide GPS-based routing, and may degrade under poor lighting conditions. Haptic feedback is currently simulated.

VIII. CONCLUSION AND FUTURE WORK

This work presented a context-adaptive wearable navigation assistant integrating YOLO-based object detection, monocular approach-speed estimation, and explainable risk-based decision making. The system emphasizes practical deployability and reduced cognitive load for unstructured

Indian environments. Future work includes integrating depth sensors, expanding datasets, and deploying physical haptic hardware.

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REFERENCES

- [1] A. Gladis *et al.*, "In-out yolo glass for visually impaired navigation," *Biomedical Signal Processing and Control*, 2024.
- [2] B. Wang *et al.*, "Yolo-od: Enhanced yolo for outdoor obstacle detection," *Sensors*, 2024.
- [3] C. Chen *et al.*, "Semantic slam with yolo for assistive grasping," *Sensors*, 2021.
- [4] D. Kuriakose *et al.*, "Deepnavi: Smartphone-based assistive navigation," *Expert Systems with Applications*, 2023.
- [5] E. Tokmurziyev *et al.*, "Llm-glasses for blind assistance," *arXiv preprint arXiv:2505.XXXX*.
- [6] F. Okolo *et al.*, "Yolov8-based assistive app with audio and haptics," *Journal of Disability Research*, 2025.
- [7] P. R. AI, "Road issues detection dataset," 2023. [Online]. Available: <https://huggingface.co/datasets/Programmer-RD-AI/road-issuesdetection-dataset>
- [8] "Road pothole images dataset." [Online]. Available: <https://datasetninja.com/road-pothole-images>
- [9] "Global navigation dataset (gnd)," 2022. [Online]. Available: <https://robotixx.github.io/GND/>
- [10] Y. Liu *et al.*, "i2nav-robot: A large-scale multi-modal indoor and outdoor navigation dataset," *arXiv preprint arXiv:2508.11485*, 2024.
- [11] A. Kumar *et al.*, "An annotated dataset for footpath damage segmentation in indian cities," *Scientific Data*, 2025.